Singapore School Kinderland Primary-6

IT (Information Technology) • Term 1

CHAPTER 1 (PAGES 5–8): DETAILED STUDY GUIDE & COMPREHENSIVE PRACTICE WORKBOOK

Notice for Students & Parents:
This guide contains **complete in-depth concept explanations** from Pages 5 to 8 of our IT book, followed by an **extensive question bank** (Multiple Choice, Blanks, Matching, True/False, Scenario Analysis, and Short Answers). It is designed to develop deep conceptual mastery of MS Access 2010 rather than just rote memorization.

PART 1: DETAILED CONCEPT EXPLANATIONS (PAGES 5 – 8)

1. Starting Microsoft Access 2010 Application

Microsoft Access 2010 is a powerful Relational Database Management System (RDBMS). To launch the application on Windows:

1. Click on the **Start button** on the Windows taskbar.
2. Move the mouse and click on **All Programs**.
3. Navigate to and click on the **Microsoft Office** folder.
4. Click on **Microsoft Office Access 2010**.

The MS Access 2010 window will appear displaying the Backstage view and Available Templates.

2. Creating a Database (Two Methods)

Method A: Using a Blank Database

- Use this when you want to design custom tables and fields from scratch.
- **Steps:** In the default window, *Blank database* is selected. Type your desired database name in the **File Name** box. By default, MS Access saves files in the Documents folder. You can click the **Browse folder icon** to change the save location. Finally, click the **Create** button.
- **Initial Table:** Access creates a default table called `Table1` in **Datasheet View** with a primary key field named `ID`. Fields can be added or renamed by double-clicking the column headers.

Method B: Using Sample Templates

- Templates are pre-built databases with pre-designed tables, forms, queries, and reports for specific tasks.
- **Steps:** Click on **Sample Templates** in the left pane → select a template (such as *Faculty*, *Students*, or *Tasks*) → enter a file name on the right pane → click **Create**.

📌 **KEY FACT FILE • File Extension:**
The default file extension for databases created in Microsoft Access 2010 is `.accdb` (Access Database). Earlier versions of MS Access (2003 and older) used `.mdb`.

3. Anatomy of the MS Access 2010 Window (7 Main Components)

Component	Detailed Function & Purpose	Screen Location
1. Title Bar	Displays the name of the currently open database and the program name. Contains the three control buttons: **Minimize**, **Maximize/Restore Down**, and **Close**.	Top of the window
2. File Menu (Backstage)	Provides access to full-screen file management tasks like New, Open, Save, Save Database As, Print, and Exit.	Top-left corner
3. Quick Access Toolbar	Provides one-click shortcuts to frequently used commands like **Save**, **Undo**, and **Redo**. Can be customized with additional commands and moved below the Ribbon.	Upper-left (above Ribbon)

Component	Detailed Function & Purpose	Screen Location
4. Ribbon	The primary command center. Contains commands organized logically into **tabs** (Home, Create, External Data, Database Tools), which are further divided into related **groups**.	Below the Title Bar
5. Navigation Pane	Lists all objects in the active database organized by category (Tables, Queries, Forms, Reports). You can double-click any object here to open it.	Left side of the screen
6. Tabbed Document Area	Displays open database objects in individual tabs. Unlike older overlapping windows, tabs keep your work area tidy and allow rapid switching.	Center workspace
7. Status Bar	Displays current status information (e.g. Ready, Num Lock) and view-switching buttons (Datasheet View vs. Design View) in the lower right corner.	Bottom of the window

4. Data Types in MS Access 2010

> Every field (column) in an Access table must have a designated **Data Type**. The data type prevents input errors by specifying what kind of data the field can store.

Data Type	Description & Rules	Storage Limit	Real-World Examples
AutoNumber	Automatically creates a unique sequential number (e.g. 1, 2, 3...) whenever a new record is added. The user cannot manually type into this field.	4 Bytes (Integer)	Admission Number, Invoice No., Roll No.
Text	Stores alphanumeric characters (letters, numbers, symbols, and spaces). Numbers in text fields are NOT used for math calculations.	Up to **255 characters**	Student Name, Street Address, Phone Number, Postal Code
Memo	Designed for large blocks of descriptive or formatted text and long notes.	Up to **65,536 characters**	Detailed Book Reviews, Medical Notes, Product Descriptions
Number	Stores numeric data intended for mathematical operations (addition, averaging, percentages). Supports both decimals and integers.	1, 2, 4, or 8 Bytes	Exam Marks, Age, Item Quantity, Product Weight
Date/Time	Stores valid calendar dates and clock times in standardized formats (e.g. Short Date, Long Date).	8 Bytes	Date of Birth, Library Book Due Date, Exam Time
Currency	Stores monetary values with up to 4 decimal places, preventing rounding errors in financial transactions.	8 Bytes	School Fees, Canteen Food Prices, Book Cost
Yes/No	Logical field storing only one of two possible values (True/False, Yes/No, or On/Off). Usually appears as a checkbox.	1 Bit	Fee Paid (Yes/No), Bus Transport User (Yes/No)

PART 2: COMPREHENSIVE PRACTICE WORKBOOK (TEST YOUR KNOWLEDGE)

Section A: Multiple Choice Questions (Circle the correct answer)

1. What is the correct sequence of steps to launch MS Access 2010 on Windows?

A) Start → Microsoft Office → All Programs → Access
B) Start → All Programs → Microsoft Office → MS Access 2010
C) Start → Control Panel → Programs → MS Access
D) Start → Documents → MS Office → Access 2010

2. Which file extension is automatically assigned to a Microsoft Access 2010 database?

A) .docx
B) .xlsx
C) .accdb
D) .pptx

3. Which component of the MS Access window displays the name of the database file?

A) Status Bar
B) Navigation Pane
C) Title Bar
D) Ribbon

4. What is the maximum number of characters that can be stored in a Text data type field?

A) 128 characters
B) 255 characters
C) 65,536 characters
D) Unlimited

5. Which of the following data types generates sequential numbers automatically without user input?

A) Number
B) Text
C) AutoNumber
D) Memo

6. When you open a blank database in MS Access 2010, what is the default name of the first table?

A) Database1
B) Table1
C) Sheet1
D) NewTable

7. The pane on the left side of the MS Access window that lists all tables, queries, and forms is the:

A) Navigation Pane
B) Quick Access Toolbar
C) Status Bar
D) Backstage View

8. If a student wants to write a 500-word essay summary in a database field, which data type should be chosen?

A) Text
B) Number
C) Memo
D) AutoNumber

9. Which toolbar contains quick shortcuts for Save, Undo, and Redo commands?

A) Status Bar
B) Quick Access Toolbar
C) Navigation Pane
D) Title Bar

10. In which view does a table open automatically when created as a blank database?

A) Design View
B) Datasheet View
C) PivotTable View
D) Layout View

Section B: Fill in the Blanks with the Correct Technical Terms

Word Bank: [.accdb | Browse | 65,536 | Navigation Pane | Sample templates | Title Bar | Number | Status Bar]

1. A ready-made database with pre-designed tables and forms can be created using _____.
2. To change the folder where your database file will be saved, click on the _____ icon.
3. The _____ is located at the very top of the window and holds the Minimize, Maximize, and Close buttons.
4. The Memo data type can hold up to _____ characters of descriptive text.
5. All tables, forms, and reports in the database are listed in the _____ on the left.
6. To store student test scores that need to be averaged, the appropriate data type is _____.
7. The file extension of an MS Access 2010 database file is _____.
8. The bar located at the very bottom of the MS Access window is the _____.

Section C: True or False (Write T or F)

1. [ ] A database in MS Access 2010 can only be created using a blank database.
2. [ ] The AutoNumber data type requires the user to type a new number for every new record.
3. [ ] The Quick Access Toolbar can be moved below the Ribbon.

4. [ ] Text fields can store up to 65,536 characters.
5. [ ] Tabbed documents allow users to switch between open tables and forms easily.
6. [ ] The File Menu contains commands like New, Open, Save, and Print.
7. [ ] Number data types can store both whole numbers and decimal values.
8. [ ] The default first field in Table1 is named "StudentName".

Section D: Real-World Scenario Challenge (Choose the Best Data Type)

For each scenario below, identify the most suitable MS Access data type:

Data Item to Store	Candidate Data Type	Why is this the best choice?
1. Student's Home Address (letters and numbers)	_____	Alphanumeric, under 255 characters
2. Total Exam Marks scored in Mathematics	_____	Numeric value used for calculation
3. Date of Admission into Grade 6	_____	Standard calendar date format
4. Detailed teacher remarks and behavioral report (800 words)	_____	Long text exceeding 255 characters
5. Unique Roll Number assigned automatically to each student	_____	Auto-generated sequential integer
6. Monthly Tuition Fee amount ($150.00)	_____	Monetary values with exact decimals

Section E: Conceptual Short Answer Questions

1. **Explain the difference between creating a database using a "Blank database" versus using "Sample templates".**

   _____

   _____

2. **Why should a student roll number be set to "AutoNumber" instead of regular "Number"?**

   _____

   _____

3. **What is the main difference between the Text data type and the Memo data type?**

   _____

   _____

4. **Name any four tabs located on the Ribbon in MS Access 2010.**

   _____

PART 3: PRACTICE WORKBOOK SOLUTIONS & EXPLANATION KEY

Section A (MCQ Solutions):
1. **B** (Start → All Programs → Microsoft Office → MS Access 2010)
2. **C** (.accdb is the Access 2010 extension)
3. **C** (Title Bar shows database name and control buttons)
4. **B** (Text data type holds up to 255 characters)
5. **C** (AutoNumber generates sequential numbers automatically)
6. **B** (Default table is titled Table1)
7. **A** (Navigation Pane lists all database objects on the left)
8. **C** (Memo is used for lengthy text exceeding 255 characters)
9. **B** (Quick Access Toolbar holds Save, Undo, Redo)
10. **B** (New blank tables open in Datasheet View)

Section B (Fill in the Blanks Solutions):
1. Sample templates • 2. Browse • 3. Title Bar • 4. 65,536 • 5. Navigation Pane • 6. Number • 7. .accdb • 8. Status Bar

Section C (True or False Solutions):
1. **False** (Can also use Sample templates)
2. **False** (AutoNumber generates numbers automatically without typing)
3. **True** (Quick Access Toolbar can be positioned below the Ribbon)
4. **False** (Text holds max 255 characters; Memo holds up to 65,536)
5. **True** (Tabbed documents allow convenient switching between objects)
6. **True** (File menu manages New, Open, Save, Print)
7. **True** (Number data type accepts both integers and decimals)
8. **False** (Default first field in Table1 is named "ID")

Section D (Scenario Data Types Solutions):
1. **Text** (Alphanumeric, under 255 characters)
2. **Number** (Values used in mathematical calculations)
3. **Date/Time** (Calendar date values)
4. **Memo** (Detailed remarks requiring over 255 characters)
5. **AutoNumber** (Auto-generated unique sequential numbers)
6. **Currency** (or Number) (Monetary values)

Section E (Short Answer Model Answers):
1. *Blank database:* Starts with an empty database where you create and design custom tables from scratch. *Sample templates:* Provides a ready-made database with pre-built tables, forms, and reports ready for use.
2. *AutoNumber benefit:* It ensures every record receives a unique, sequential number automatically, preventing human typing errors and duplicate IDs.
3. *Text vs. Memo:* Text holds short alphanumeric data up to 255 characters, while Memo holds large blocks of text up to 65,536 characters.
4. *Ribbon Tabs:* Home, Create, External Data, Database Tools.